

Generation Interconnection Request System Impact Study Report

For Queue Project Q19-1337

East Bismarck 100 kV

October 17, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1337. The Interconnection Customer proposes a 231.2 MW solar photovoltaic facility interconnecting at the East Bismarck 100 kV substation in the Illinois area. The request was received on November 15, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 356 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 7 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$13,557,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Grand Coulee Number One - Sunol Tap 100 kV
Contingency: Texon Substation - E Bismarck 230 kV
Loading: 121.3% (Rating: 125 MVA)
DFAX: 20.4%
Pre-project loading: 83.6%
2. Facility: Mtn Island - Terminous Substation 100 kV L
Contingency: East Bismarck - Texon Substation 230 kV line
Loading: 119.7% (Rating: 183 MVA)
DFAX: 37.8%
Pre-project loading: 72.0%
3. Facility: Rio Bravo Rocklin - East Bismarck 100 kV L
Contingency: Texon - E Bismarck 230 kV
Loading: 117.8% (Rating: 189 MVA)
DFAX: 26.7%
Pre-project loading: 85.2%
4. Facility: E Bismarck - Rio Bravo Rocklin 100 kV #2
Contingency: Texon - East Bismarck 230 kV
Loading: 117.8% (Rating: 189 MVA)
DFAX: 26.7%
Pre-project loading: 85.2%
5. Facility: Terminous - Lake Jackson 100 kV
Contingency: Texon - East Bismarck Substation 230 kV line
Loading: 115.2% (Rating: 217 MVA)
DFAX: 37.8%
Pre-project loading: 74.8%

6. Facility: Lake Jackson - E Bismarck 100 kV
 Contingency: East Bismarck - Texon 230 kV Line
 Loading: 112.5% (Rating: 164 MVA)
 DFAX: 25.8%
 Pre-project loading: 76.2%
7. Facility: Rio Bravo Rocklin - Sunol Ap 100 kV
 Contingency: E Bismarck - Texon 230 kV
 Loading: 103.9% (Rating: 203 MVA)
 DFAX: 34.7%
 Pre-project loading: 64.3%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Grand Coulee Number One - Sunol Tap 100 kV	Rebuild line with higher-capacity conductor	\$3,472,000	50% (\$1,724,000)
Mtn Island - Terminous Substation 100 kV L	Rebuild line with higher-capacity conductor	\$2,201,000	50% (\$1,093,000)
Rio Bravo Rocklin - E Bismarck 100 kV	Rebuild line with higher-capacity conductor	\$7,872,000	50% (\$3,910,000)
E Bismarck - Rio Bravo Rocklin 100 kV #2	Rebuild line with higher-capacity conductor	\$7,548,000	50% (\$3,748,000)
Lake Jackson - Terminos 100 kV Line	Rebuild line with higher-capacity conductor	\$1,145,000	50% (\$569,000)
E Bismarck - Lake Jackson 100 kV	Rebuild line with higher-capacity conductor	\$3,715,000	50% (\$1,846,000)
Rio Bravo Rocklin - Sunol Tap 100 kV L	Reconductor line	\$1,343,000	50% (\$667,000)
	Total allocated to project		\$13,557,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.